

CHAPTER 10

Plasmons and Magnons

Contents

- ❑ In this chapter we will briefly address some other collective excitations in semiconductors and the quasi-particles which result from the quantization of these excitations like plasmons or magnons
- ❑ Considering a semiconductor which contains a large number of electrons ($10^{17} - 10^{19} \text{cm}^{-3}$), collective oscillations of electrons → Plasma oscillations
- ❑ Magnons, existing in ferro-, ferri- and antiferromagnetic materials, the collective excitations of the spin system

10.1 Plasmons, Pair Excitations and Plasmon-Phonon Mixed States

A surface charge density: $\rho_s = ne\Delta x$

The electric field $E_x = \frac{ne\Delta x}{\epsilon\epsilon_0}$

The electric field acts back on the electrons

$$-eE_x = -\frac{ne^2\Delta x}{\epsilon\epsilon_0} = m_e \frac{\partial^2}{\partial t^2} (\Delta x)$$

Which is a harmonic oscillation with frequency:

$$\omega_{PL}^0 = \left(\frac{ne^2}{m_e\epsilon\epsilon_0} \right)^{\frac{1}{2}} = \omega_L, \omega_T = 0 \quad k \rightarrow 0$$

If we go to shorter wavelengths we find a weak parabolic dispersion for the plasmons

$$\omega_{PL}(k) = \omega_{PL}^0(1 + ak^2 + \dots)$$

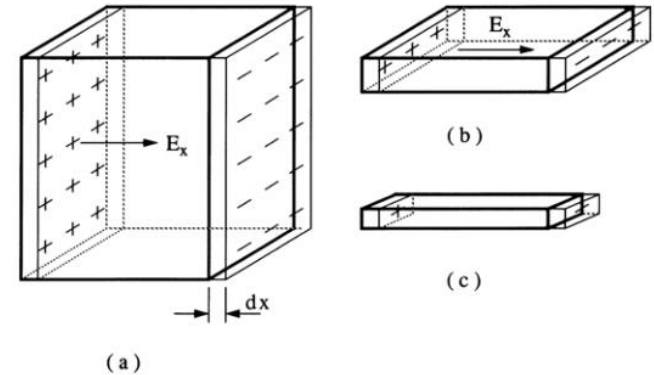


Fig. 10.1. Schematic drawing of a displacement of the electron system in a plasma oscillation in a three- (a), two- (b) and one-dimensional semiconductor (c)

Plasma oscillation-electron collective oscillation

- Only longitudinal oscillations can propagate
- The quanta of plasma oscillation is plasmon
- The plasmons obey Bose statistics
- For large wavevectors: the plasmon modes are strongly damped
- For semiconductors, the plasmon energy is 10-100 meV
- For metals, the plasma frequency is usually in Vis-UV range

The dielectric function of plasmons:

$$\varepsilon(\omega) = 1 + \frac{Ne^2}{\varepsilon_0 m} (\omega'^2_0 - \omega^2 - i\omega\gamma)^{-1}$$

$$\varepsilon(\omega) = \varepsilon_b + \frac{f_{j'}}{\omega^2_{0j'} - \omega^2 - i\omega\gamma_{j'}}$$



$$\varepsilon(\omega) = \varepsilon_b + \frac{\frac{ne^2}{m_e \varepsilon_0}}{-\omega^2 - i\omega\gamma} \quad (10.3)$$

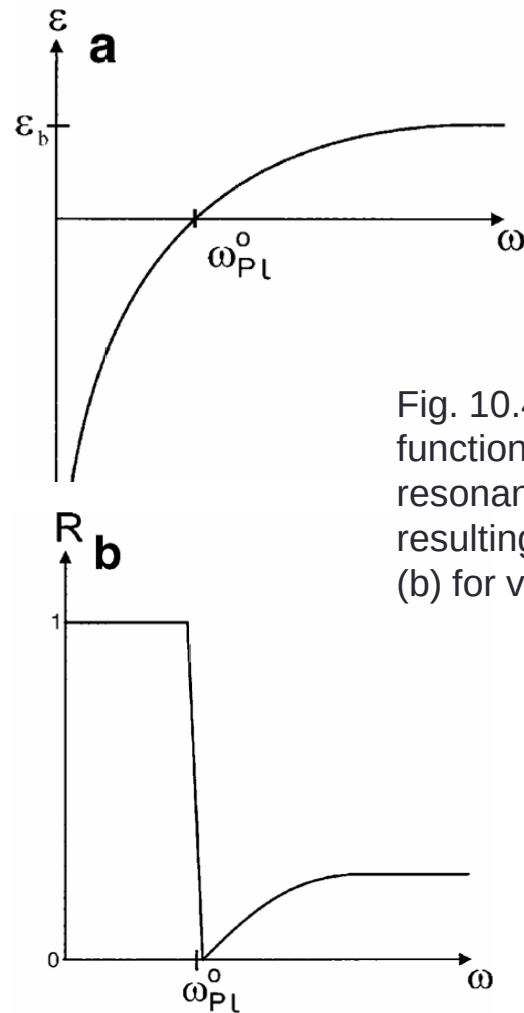
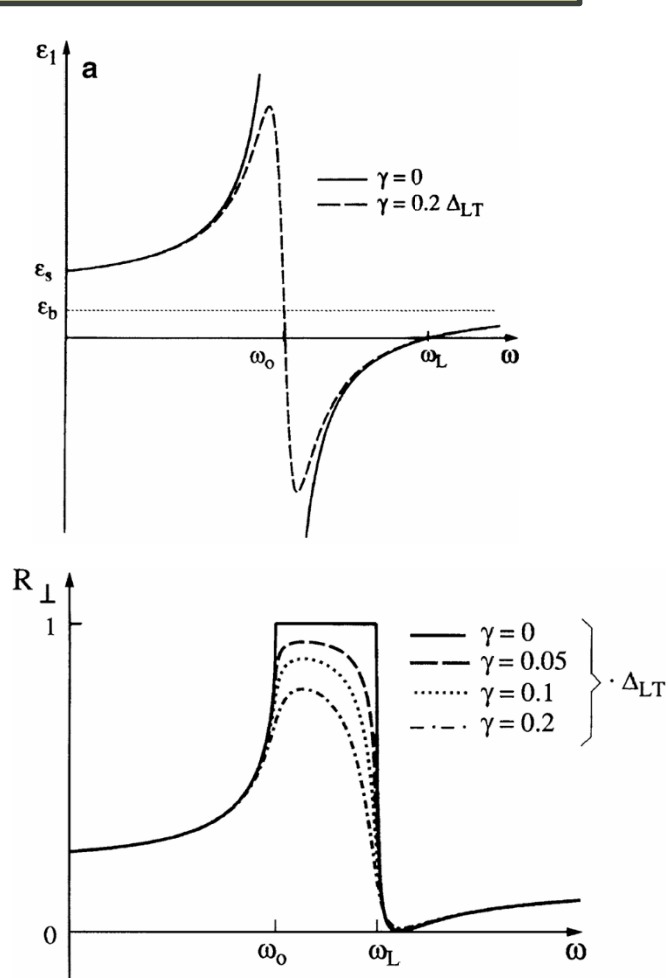


Fig. 10.4 The dielectric function of a plasmon resonance (a) and the resulting reflection spectrum (b) for vanishing damping

The equation of motion for free carriers for an electric field :

$$m\ddot{x} = F_x = -\frac{m}{\tau}\dot{x} + eE_0^0 e^{-i\omega t} \quad m\ddot{x} + \gamma m\dot{x} + \beta x = eE_0 e^{-i\omega t} \quad (10.4a)$$

Resulting in a frequency-dependent electrical conductivity:

$$\sigma(\omega) = \sigma_0 \frac{1}{1 - i\omega\tau} \quad (10.4b)$$

and to a total dielectric function:

$$\varepsilon_{total}(\omega) = \varepsilon_{Lorentz}(\omega) + \chi_{free\ carrier}(\omega) \quad (10.4c)$$

with the help of Maxwells equation $\nabla \times \mathbf{H} = \mathbf{j} + (\varepsilon_0 \dot{\mathbf{E}} + \dot{\mathbf{P}})$ $\nabla \times \mathbf{H} = \mathbf{j} + \dot{\mathbf{D}} = \sigma \mathbf{E} + \varepsilon_0 \varepsilon_{Lorentz} \dot{\mathbf{E}}$
we obtain : $\mathbf{P} = \varepsilon_0 \chi \mathbf{E}$

$$\chi_{free\ carrier}(\omega) = \frac{i}{\varepsilon_0 \omega} \sigma(\omega)$$

and for the total dielectric function:

$$\varepsilon(\omega) = \varepsilon_b + \frac{\frac{ne^2}{m\varepsilon_0}}{-\omega^2 - i\omega\gamma}$$

$$\varepsilon_{total}(\omega) = \varepsilon_{Lorentz}(\omega) + \frac{i}{\varepsilon_0 \omega} \sigma(\omega) = \varepsilon_{Lorentz}(\omega) + \frac{\omega_{PL}^2}{-\omega^2 - i\omega/\tau}$$

Quantized (surface) plasmon modes exist in small metal spheres or colloids.

史上最诡异的古罗马文物: Lycurgus变色玻璃杯 (藏于大英博物馆)

- 根据大英博物馆官方资料, 此杯的制作方法是一体成形, 先吹制厚玻璃, 再逐点削去形成突出浮雕图案. 由於玻璃(硬度7)不能用金属工具有效切削, 只能通过钻石(10)或蓝宝石(8-9)切削, 其加工难度可见一斑.
- 左半边是外光源, 玻璃呈暗绿色; 右半边是内光源(如将灯泡放置里面), 玻璃竟然通体散发出诡异的血红色. 杯子高16.5公分, 直径13.2公分, 制作於四世纪, 生产地点相信是罗马城或是亚历山大城.



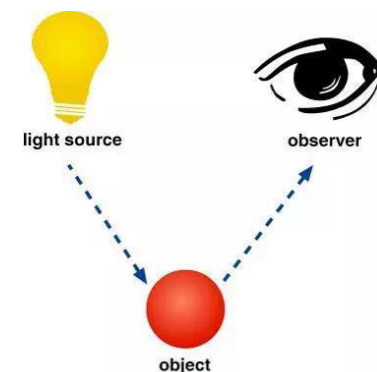
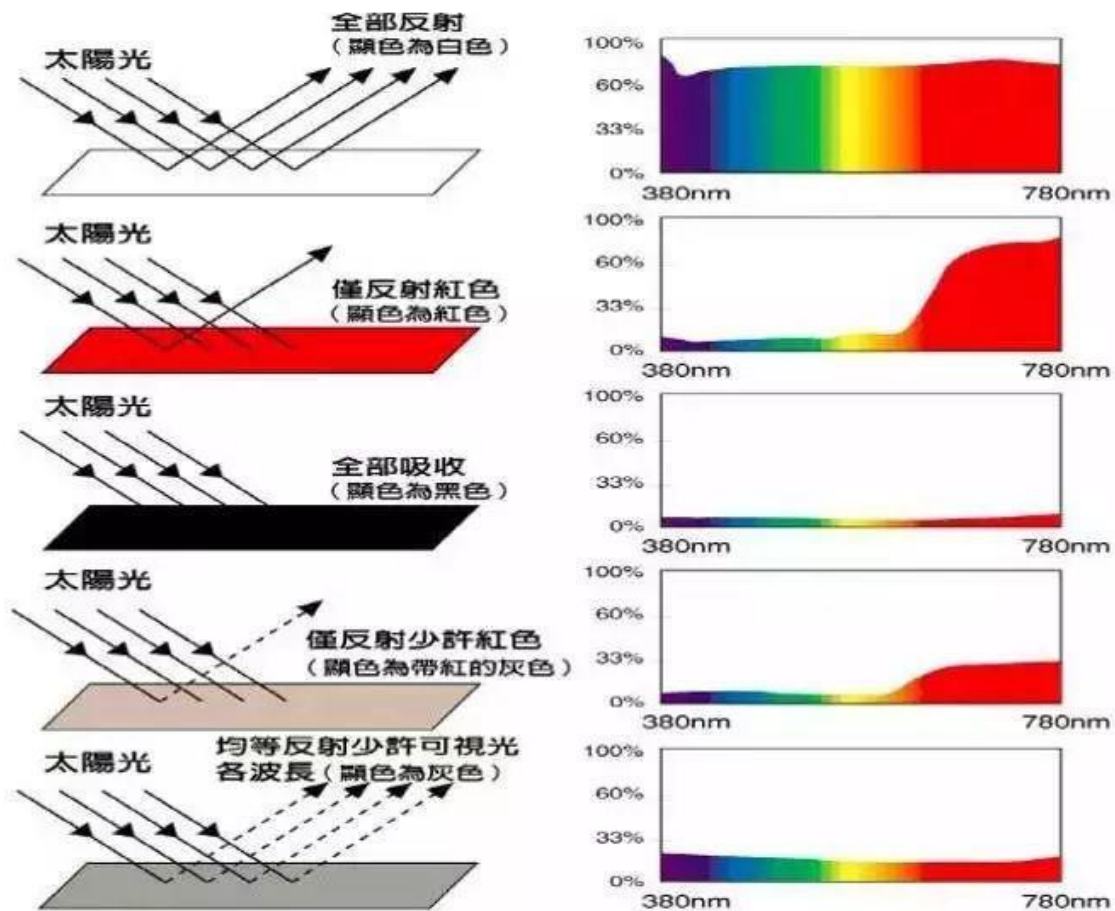
经现代技术鉴证测试, 玻璃含有40ppm黄金和330ppm白银, 金银在制造玻璃时混合成约70纳米直径微粒, 表面等离子体共振(surface plasmon resonance)现象解释了奇特的光学效果。

http://www.britishmuseum.org/explore/highlights/highlight_objects/pe_mla/t/the_lycurgus_cup.aspx

胭脂红瓷器



光谱反射和颜色感知



10.2 Magnons and Magnetic Polarons

- Magnons, existing in ferro-, ferri- and antiferromagnetic materials, the collective excitations of the spin system.
- Magnons: The quanta of a collective excitation, which is the result of a ferromagnetic spin system would be to tilt one spin by an amount $\hbar$ off the magnetization direction.
- In antiferromagnetic or ferrimagnetic materials :
 - The lowest branch is called the “acoustic” magnon branch ;
 - The higher ones “optic” magnon branches .
- Magnetic polarons :
 - $\text{Cd}_{1-x}\text{Mn}_x\text{Te}$ 、 $\text{Zn}_{1-x}\text{Mn}_x\text{Se}$ 、 $\text{Ga}_{1-x}\text{Mn}_x\text{As}$.
 - The electron can still move freely through the lattice, but it carries with it a magnetization cloud.

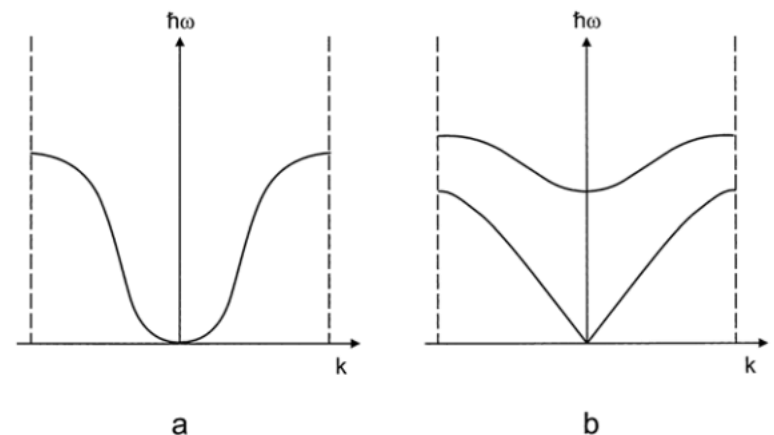


Fig. 10.7. The schematic dispersion relation of magnons in a ferromagnet (a) and an antiferromagnet (b)